

CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATED ASSIGNMENT SYSTEM

Using Salesforce and Agentforce

Project Documentation

Custom Object: Support Ticket Intelligence
Flow: Support_Ticket_Intelligence – Auto-Launched Flow
Agentforce Subagent: Support Ticket Priority Analysis

Prepared for Salesforce / Agentforce project implementation

College Name: University College of Engineering Ramanathapuram

College Code:9130

Semester :07

Team ID : SWTID-2026-7490

Team Size : 4

Team Leader : Rajesh Kumar R M

Team member : Surya R

Team member : Arunkumar A

Team member : Logesh G

1. Executive Summary

The Customer Support Ticket Priority Prediction and Automated Assignment System is a Salesforce-based solution designed to reduce manual effort in customer support operations. The system stores customer support requests in a custom Salesforce object, analyzes the latest ticket associated with a customer Account, classifies the ticket as High, Medium, or Low priority, assigns an appropriate support level, and creates an urgent handling task for High-priority tickets. Agentforce provides the conversational AI layer while Salesforce Flow performs the backend automation.

The documented solution uses the Support Ticket Intelligence custom object, an Auto-Launched Flow, and an Agentforce subagent named Support Ticket Priority Analysis. The intended outcome is faster identification of critical tickets, reduced repetitive work, more consistent support handling, and improved awareness of SLA-related risk.

2. Project Objective

The objective is to understand how customer support teams manage large volumes of service requests and the challenges involved in manual ticket prioritization and assignment. The project implements an intelligent Salesforce and Agentforce solution that analyzes ticket details, determines priority, assigns support responsibility, and triggers additional automation for critical cases.

- Centralize creation, storage, and management of customer support tickets.
- Automatically classify tickets as High, Medium, or Low priority.
- Use Agentforce to analyze ticket information and trigger backend automation.
- Assign tickets to the appropriate support level or agent.
- Create an urgent handling task for High-priority tickets.
- Provide support managers with visibility into ticket status, priority, assignment, and SLA risk.
- Improve response and resolution efficiency while reducing manual prioritization work.
- Maintain controlled access to customer and ticket information.

3. Problem Statement

Customer support teams may receive many tickets while relying on manual review to identify urgent requests. Important urgency signals can be missed, critical issues may wait behind lower-priority requests, and assignment decisions can require manual effort. Managers also need better visibility into older unresolved tickets and potential SLA risk.

Persona	Need	Current Challenge	Desired Gain
Support Agent	Focus on critical tickets first	Manual prioritization and missed urgency signals	Automatic priority classification and clearer assignment
Customer	Get urgent issues resolved quickly	Critical issues may wait behind lower-priority tickets	Faster identification and handling of urgent cases
Support Manager	Assign tickets to the right support level	Manual assignment decisions	More consistent support assignment
Support Manager	Reduce SLA-related risk	Older unresolved tickets may remain unnoticed	SLA risk awareness

4. Approach and Requirement Gathering

The project approach begins by understanding the existing support-ticket process, including how tickets are created, prioritized, assigned, and resolved. Stakeholder needs include support agents, support managers, and customers. Sample ticket descriptions are analyzed for recurring urgency indicators such as “urgent”, “not working”, “failure”, “issue”, “slow”, and “delay”.

1. Gather requirements from support agents, managers, and customers.
2. Study the current ticket creation, prioritization, and assignment process.
3. Identify pain points such as inconsistent prioritization, delays, and uneven workload.
4. Analyze sample ticket descriptions for common urgency indicators.
5. Design the Salesforce data model and automation.
6. Configure the Auto-Launched Flow for ticket retrieval, classification, assignment, and task creation.
7. Configure Agentforce to provide conversational access to the ticket-priority process.
8. Test High, Medium, and Low priority scenarios and verify the resulting automation.

5. Functional Requirements

ID	Requirement
FR-1	Support ticket creation and management
FR-2	Account and Contact association
FR-3	Ticket description analysis
FR-4	Automatic priority classification
FR-5	High-priority task creation
FR-6	Support agent assignment
FR-7	SLA breach risk checking
FR-8	Agentforce conversational analysis

6. Non-Functional Requirements

Category	Requirement
Usability	Clear Agentforce interaction and understandable priority messages.
Reliability	Consistent priority classification based on configured conditions.
Performance	Quick retrieval and analysis of the latest support ticket.
Security	Controlled Salesforce access to customer and ticket information.
Maintainability	Flow and Agentforce configuration can be updated as requirements change.
Scalability	Support increasing ticket volumes through Salesforce automation.

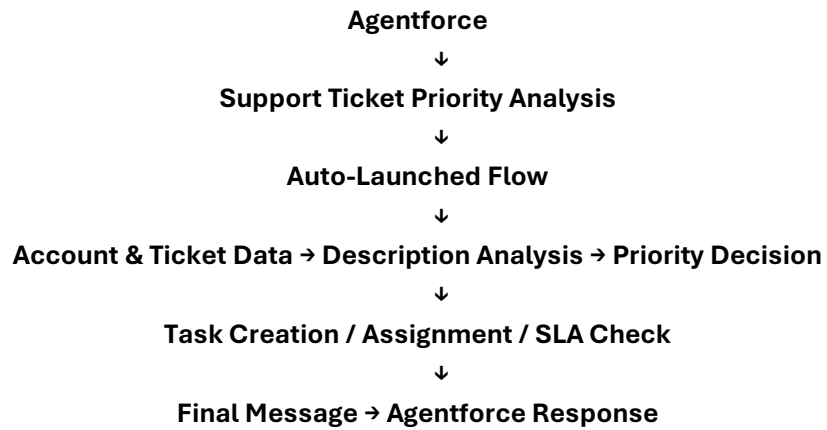
7. Solution Architecture

High-level architecture:

Support User / Support Team



Customer Support Ticket Priority Prediction and Automated Assignment System | Salesforce + Agentforce



The architecture separates conversational interaction from backend record processing. Agentforce handles the user-facing analysis experience, while Flow performs record retrieval, decision logic, task creation, assignment, and final output preparation.

8. Technology Stack

Layer	Technology / Component	Purpose
Platform	Salesforce	CRM platform and secure data storage
Database	Support_Ticket_Intelligence__c	Stores support ticket information
Related Records	Account, Contact, User, Task	Customer context, contact, assignment, and urgent work
Logic	Auto-Launched Flow	Retrieves, analyzes, classifies, and automates ticket handling
AI	Agentforce	Conversational ticket analysis and automation entry point
Agent Configuration	Support Ticket Priority Analysis	Analyzes ticket descriptions and determines priority
Automation	Decision, Assignment, Create Records	Implements priority, assignment, and task behavior

9. Salesforce Data Model

The core custom object is Support Ticket Intelligence. The documented API name is Support_Ticket_Intelligence__c.

Field Label	API Name	Data Type	Documented Values / Format	Purpose
Ticket Number	Ticket_Number__c	Auto Number	TKT-{0000}	Unique ticket identifier
Customer	Customer__c	Lookup (Account)	Account	Related customer account
Contact	Contact__c	Lookup (Contact)	Contact	Customer contact
Issue Type	Issue_Type__c	Picklist	Technical, Billing, General	Categorizes issue
Description	Description__c	Long Text Area	Free text	Stores issue details
Priority Level	Priority_Level__c	Picklist	Low, Medium, High	Calculated ticket priority
Status	Status__c	Picklist	New, In Progress,	Tracks ticket state

			Resolved	
Created Date	Created_Date__c	Date	Date	Ticket date
Assigned To	Assigned_To__c	Lookup (User)	User	Support agent
SLA Breach Risk	SLA_Breach_Risk__c	Checkbox	True / False	Risk flag
Resolution Time (hrs)	Resolution_Time__c	Number	Numeric hours	Time taken to resolve

Note: The table above follows the field definitions documented in the provided project PDF. If the Salesforce org currently uses a different field type or label, the actual org configuration should be recorded separately in the final implementation evidence.

The screenshot shows the Salesforce Setup interface for the 'Support Ticket Intelligence' object. The 'Fields & Relationships' section is active, displaying a list of 15 items sorted by field label. The fields shown are:

Field Name	Field Label	Field Type	Required
Priority Level	Priority_Level__c	Picklist	
Resolution Time	Resolution_Time__c	Number(18, 0)	
SLA Breach Risk	SLA_Breach_Risk__c	Checkbox	
status	status__c	Picklist	
Ticket Number	Name	Auto Number	✓
User	User__c	Lookup(User)	✓

10. Ticket Lifecycle

9. A customer support request is represented as a Support Ticket Intelligence record.
10. The ticket is associated with an Account and, where available, a Contact.
11. The system retrieves the latest ticket associated with the Account.
12. The Description field is analyzed for configured urgency indicators.
13. The ticket is classified as High, Medium, or Low.
14. High-priority tickets trigger urgent handling automation.
15. The ticket is assigned to an appropriate support level, with High-priority cases assigned to a senior support agent in the documented design.
16. An optional SLA-risk check can identify older unresolved tickets.
17. A clear result is returned to the Agentforce user.
18. The support team continues normal handling until the ticket is resolved.

11. Priority Classification Logic

The documented implementation uses keyword-based decision logic. This is a configured business rule, not an unrestricted AI judgment.

Priority	Configured Indicators	Documented Handling
High	urgent; not working; failure	High-priority handling; create urgent task; assign senior support level
Medium	issue; slow; delay	Mark Medium and continue normal support handling
Low	None of the above	Mark Low and queue for processing

The keyword logic is case-dependent on the configured Flow/Agentforce implementation. The documented design recommends using CONTAINS-style conditions for the decision.

12. Auto-Launched Flow Design

The project uses an Auto-Launched Flow with no record trigger. It is intended to be called from Agentforce and can also be tested independently in Flow Builder.

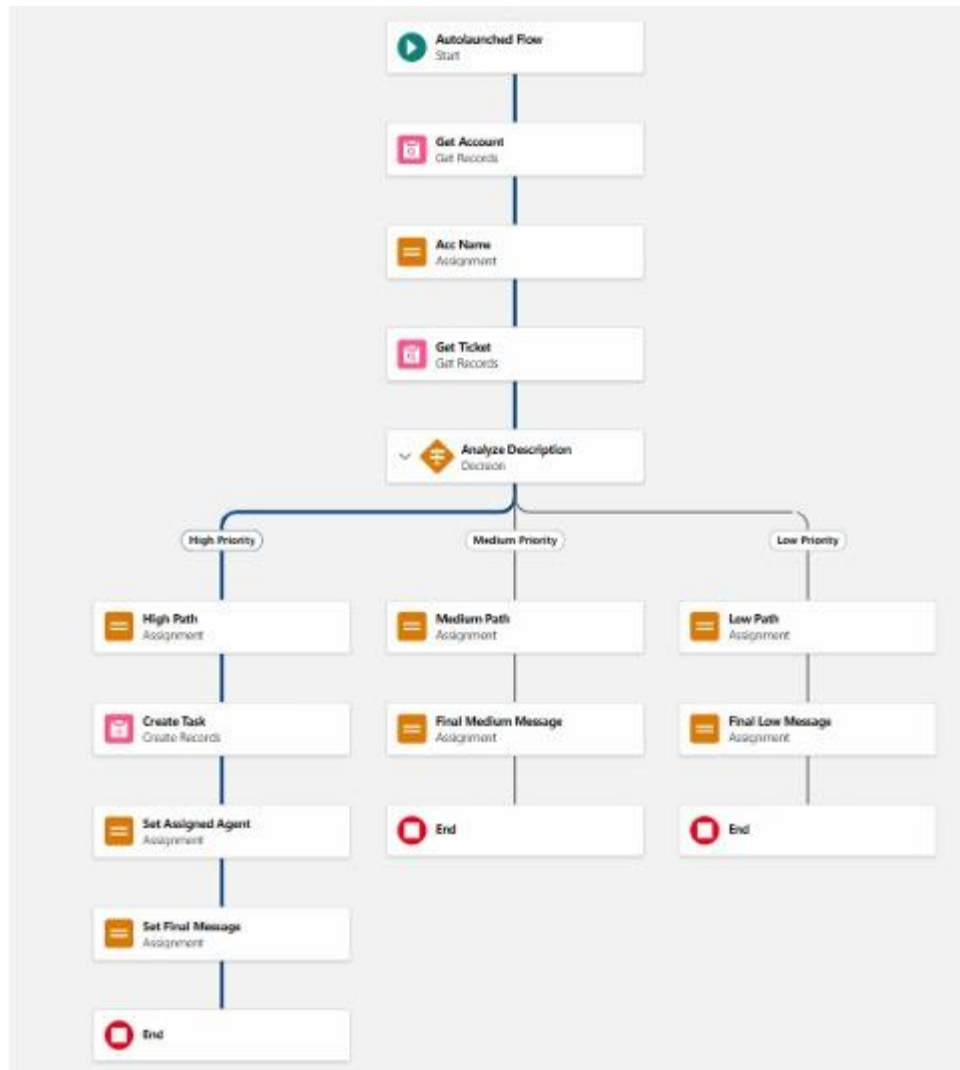
12.1 Flow Variables

Variable	Type	Direction	Purpose
varAccountName	Text	Input	Account name supplied by the user
varAccountId	Text	Output	ID of the matching Account
varTicketId	Text	Output	ID of the latest support ticket
varPriorityLevel	Text	Output	High, Medium, or Low
varAssignedTo	Text	Output	Support level / agent assignment
varActionMessage	Text	Output	Final response message

12.2 Flow Steps

19. Get Account — Get the Account record where Name equals varAccountName; sort by CreatedDate descending and store only the first record.
20. Store Account Id — Assign Get_Account.Id to varAccountId.
21. Get Ticket — Query Support_Ticket_Intelligence__c where Customer__c equals varAccountId; sort by CreatedDate descending and store only the first record.
22. Store Ticket Id — Assign Get_Ticket.Id to varTicketId.
23. Analyze Description — Use a Decision element to classify the Description.
24. Set Priority Level — Set varPriorityLevel to High, Medium, or Low depending on the selected path.
25. Is High Priority — Check whether varPriorityLevel equals High.
26. Create Urgent Task — For High priority, create a Task with Subject 'Urgent Ticket Handling', WhatId = varTicketId, Priority = High, and Status = Not Started.
27. Set Assigned Agent — Set varAssignedTo to the appropriate support level; the documented example uses 'Senior Support Agent' for High priority.
28. SLA Risk — Optionally flag tickets whose Created Date is older than two days.
29. Set Final Message — Prepare the appropriate High, Medium, or Low response.
30. End — Return the configured outputs to the Agentforce interaction.

12.3 Flow Structure



**Start → Get Account → Store Account Id → Get Ticket → Store Ticket Id
→ Analyze Description → Set Priority Level → Is High Priority
→ Create Task (High only) → Set Assigned Agent → Optional SLA Decision
→ Set Final Message → End**

13. Agentforce Configuration

The documented Agentforce component is a subagent named Support Ticket Priority Analysis with API name Support_Ticket_Priority_Analysis.

13.1 Classification / Scope

The topic analyzes customer support tickets based on issue description and urgency. Its scope is limited to ticket analysis, priority determination, and backend automation for task assignment. The documented scope excludes unrelated requests such as billing, subscription management, or account updates.

13.2 Agent Instructions

31. Ask the user for the Account Name if it is not already available.
32. Retrieve the latest support ticket associated with the Account.
33. Read and analyze the ticket Description.
34. Determine High, Medium, or Low priority using the configured indicators.
35. Set the priority level accordingly.
36. For High priority, trigger the Auto-Launched Flow and ensure an immediate-handling task is created.
37. Assign the ticket to the appropriate support level, such as a senior agent for High priority.
38. Return a clear response indicating whether urgent handling was initiated, normal handling will occur shortly, or the ticket is queued.
39. Do not update records directly or send emails from the Agentforce topic; use Flow for backend actions.

13.3 Agent Inputs and Outputs

Name	Type	Direction	Purpose
varAccountName	lightning__textType / Text	Input	Account name used to find the latest ticket
varAccountId	lightning__textType / Text	Output	Account identifier
varTicketId	lightning__textType / Text	Output	Analyzed ticket identifier
varPriorityLevel	lightning__textType / Text	Output	Calculated priority
varAssignedTo	lightning__textType / Text	Output	Assigned support agent/team
varActionMessage	lightning__textType / Text	Output	Final conversational result

14. Agentforce-to-Flow Interaction

The intended interaction is conversational. A support user provides an Account Name. Agentforce invokes the Support Ticket Priority Analysis topic, which calls the backend Flow. The Flow retrieves the latest ticket, analyzes its Description, determines priority, performs High-priority automation, and returns outputs that Agentforce can present to the user.

**User → Account Name → Agentforce → Support Ticket Priority Analysis
→ Auto-Launched Flow → Latest Ticket → Priority / Assignment / Task
→ Agentforce Response**

15. Final Response Messages

Priority	Documented Response
High	High priority ticket detected. Assigned to senior agent.
Medium	Ticket marked as medium priority. Will be handled shortly.
Low	Tickets are low priority and queued for processing.

16. High-Priority Task Automation

When the ticket is classified as High, the Flow creates a Salesforce Task. The documented Task settings are:

Task Field	Value
Subject	Urgent Ticket Handling

WhatId	The analyzed Support Ticket Intelligence record
Priority	High
Status	Not Started

17. SLA Breach Risk

The project includes an optional SLA-risk check. The documented example flags a ticket when its Created Date is older than two days. The current design is an awareness check rather than a complete SLA monitoring and escalation platform.

18. Security and Access

- Use Salesforce permissions to control access to Accounts, Contacts, Users, Tasks, and Support Ticket Intelligence records.
- Restrict sensitive customer and ticket data according to organizational roles.
- Support agents should receive only the access needed to work their tickets.
- Managers require broader visibility for monitoring ticket status, priority, assignment, and workload.
- Agentforce actions should use the configured Flow rather than allowing the conversational layer to bypass business logic.

19. Testing Plan

The following test cases provide a practical validation matrix for the documented design.

Test Case	Description	Expected Result
TC-01	Description contains 'urgent'	Priority = High; urgent task created
TC-02	Description contains 'not working'	Priority = High; urgent task created
TC-03	Description contains 'failure'	Priority = High; urgent task created
TC-04	Description contains 'issue'	Priority = Medium
TC-05	Description contains 'slow'	Priority = Medium
TC-06	Description contains 'delay'	Priority = Medium
TC-07	Description contains none of the configured keywords	Priority = Low
TC-08	High-priority ticket	Assigned support level should be senior support; Task created
TC-09	Older unresolved ticket	Optional SLA risk check should flag the ticket according to configuration
TC-10	Agentforce conversation with Account Name	Agentforce retrieves latest ticket and returns the configured result

20. Sample Test Ticket

A sample High-priority test ticket can be created with the following values:

Field	Sample Value
Customer	An existing Salesforce Account
Contact	A Contact associated with that Account
Issue Type	Technical

Description	The customer system is not working. The service has completely failed and the issue is urgent.
Priority Level	Expected: High
Status	New
Assigned To	Expected: Senior Support Agent / configured support level
SLA Breach Risk	Depends on configured SLA logic
Resolution Time (hrs)	Leave blank until resolved

The screenshot displays the Salesforce Developer Edition interface. At the top, there's a navigation bar with a search bar and various utility icons. Below this, the main content area shows a 'Support Ticket Intelligence' record for 'TKT-0000'. The record details are as follows:

- Issue Type: 1
- Description: hi
- Priority Level: (blank)
- Customer: rajesh kumar
- User: EinsteinServiceAgent User
- SLA Breach Risk: (blank)
- Resolution Time: (blank)
- status: (blank)

On the right side of the record, there are buttons for 'New Contact', 'Edit', and 'New Opportunity'. The left sidebar shows the 'Developer Edition' logo and a 'Welcome' message.

21. Advantages

- Automatic ticket prioritization.
- Faster identification of critical issues.
- Reduced manual effort for prioritization, assignment, and urgent task creation.
- Agentforce-assisted ticket analysis.
- Automatic urgent task creation for High-priority tickets.
- Improved support assignment consistency.
- Better SLA-risk awareness.
- Scalability for increasing support-ticket volumes.
- Improved team productivity by reducing repetitive work.

22. Limitations and Risks

- Keyword-based priority logic can miss urgency that is expressed without configured keywords.
- The documented implementation has limited contextual understanding because priority is based on configured conditions.
- Correct Account, Contact, and ticket information are required.
- Flow configuration must be maintained when ticket-handling rules change.
- Agentforce behavior depends on correctly configured instructions, topics, actions, and Flow integration.

- The documented Agentforce scope is limited to ticket analysis, priority determination, and backend automation.
- SLA monitoring is limited to an optional risk check rather than full escalation management.
- More advanced assignment rules would require additional configuration.
- Advanced analytics for priority trends and resolution performance are identified as future scope.

23. Future Scope

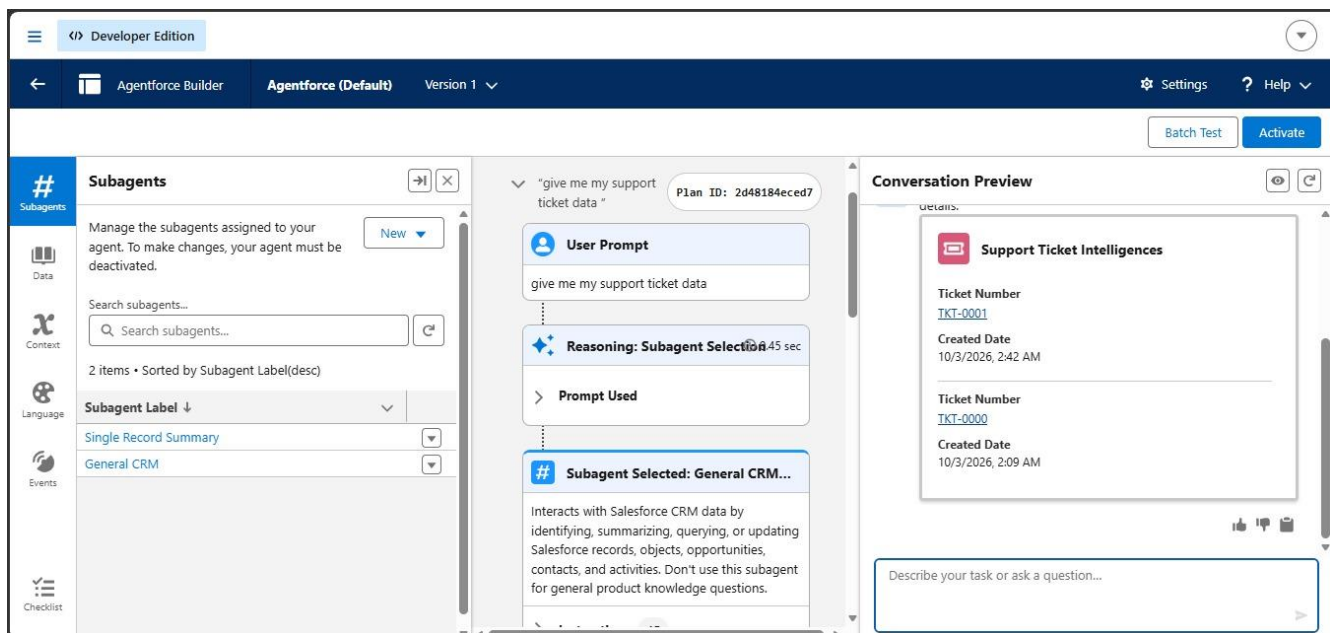
- Add more priority indicators and business rules.
- Extend Agentforce outputs with additional ticket information.
- Implement more sophisticated SLA monitoring and escalation.
- Support additional support teams and assignment rules.
- Introduce analytics for priority trends and resolution performance.
- Add more Flow variables and Agentforce action parameters for richer outputs.
- Consider broader contextual analysis beyond keyword matching.
- Add management dashboards for workload, priority distribution, SLA risk, and resolution time.

24. Implementation Checklist

Area	Implementation Item	Status for Final Submission
Data Model	Support Ticket Intelligence custom object created	Verify in org
Data Model	Ticket Number Auto Number TKT-{0000}	Verify in org
Data Model	Customer lookup to Account	Verify in org
Data Model	Contact lookup to Contact	Verify in org
Data Model	Issue Type picklist	Verify in org
Data Model	Description long text	Verify in org
Data Model	Priority Level picklist	Verify in org
Data Model	Status picklist	Verify in org
Data Model	Assigned To lookup to User	Verify in org
Data Model	SLA Breach Risk checkbox	Verify in org
Data Model	Resolution Time number	Verify in org
UI	Custom object tab accessible	Completed in current implementation
Testing	Sample ticket can be created	Verify with test record
Flow	Auto-Launched Flow configured	Complete / verify
Flow	Get Account and latest Ticket logic	Complete / verify
Flow	Priority Decision logic	Complete / verify
Flow	High-priority Task creation	Complete / verify
Agentforce	Support Ticket Priority Analysis created	Complete / verify
Agentforce	Flow action connected	Complete / verify
Testing	High / Medium / Low test cases passed	Complete / capture evidence
Reporting	Manager dashboard / reports	Optional / project scope dependent

25. Recommended Evidence for Project Submission

40. Screenshot of the Support Ticket Intelligence object in Object Manager.
41. Screenshot of Fields & Relationships showing the configured fields.
42. Screenshot of the Support Ticket Intelligences tab showing a created test ticket.
43. Screenshot of the Auto-Launched Flow canvas showing the complete flow structure.
44. Screenshot of the Flow variables and Decision conditions.
45. Screenshot of the Agentforce Support Ticket Priority Analysis configuration.
46. Screenshot of the Agentforce action connected to the Flow.
47. Screenshot showing a High-priority ticket and the resulting Urgent Ticket Handling Task.
48. Screenshots of Medium and Low test cases.
49. Screenshot of the final Agentforce response.
50. Screenshots of any reports or dashboard created for managers.



26. Conclusion

The Customer Support Ticket Priority Prediction and Automated Assignment System demonstrates how Salesforce Flow and Agentforce can be combined to reduce manual support-ticket handling. The documented solution retrieves the latest ticket for an Account, analyzes its Description, classifies priority, creates an urgent task for High-priority cases, assigns an appropriate support level, and returns a clear conversational response. The design supports faster handling of critical tickets, reduced manual work, intelligent prioritization, and improved support-team productivity.

27. Appendix A — Key Salesforce Names

Component	Documented Name
Custom Object Label	Support Ticket Intelligence
Custom Object API Name	Support_Ticket_Intelligence__c
Flow	Support_Ticket_Intelligence – Auto-Launched Flow
Agentforce Subagent	Support Ticket Priority Analysis

Agentforce API Name	Support_Ticket_Priority_Analysis
Primary Related Records	Account, Contact, User, Task

28. Appendix B — Source Basis and Scope Note

This document was prepared from the project material provided in the conversation, including the PDF titled 'Customer Support Ticket Priority Prediction and Automated Assignment System Using Agentforce'. Terminology, field definitions, priority keywords, Flow structure, Agentforce configuration, advantages, limitations, conclusion, and future scope have been preserved from that source where applicable.

Where the Salesforce org may differ from the documented design, the actual org configuration should be used as the final implementation record. This is particularly important for field data types, Flow variables, Agentforce UI options, and the exact current status of configuration.